Hybrid Neuro-Fuzzy Systems

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Summary

- Neuro-Fuzzy Systems
- Structures of Neuro-Fuzzy Systems
- Case study : FDI OF AN ELECTRO - PNEUMATIC VALVE ACTUATOR

Neuro-Fuzzy Systems

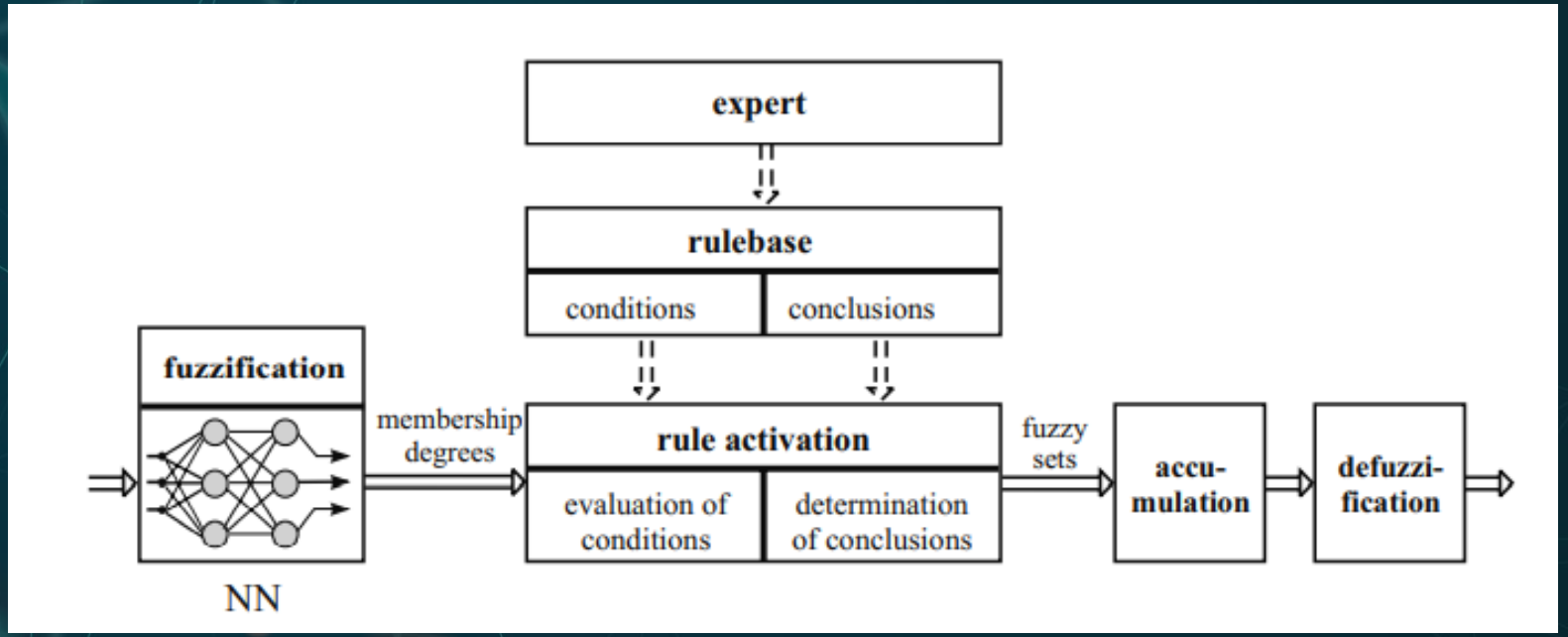
- Neural Network training techniques handles information retrieval from data using optimization techniques
- Fuzzy system provides the intuitive understanding of the resulting system and establishes the possibility of integrating expert knowledge
- Neuro-Fuzzy Systems is a combination of **“Human experts and Experimental data”**
- It can generate new rules from the data and also refine the existing rules by adapting parameters within them

Application Areas

01 Control - Approximate the non-linear functions in modelling and system identification

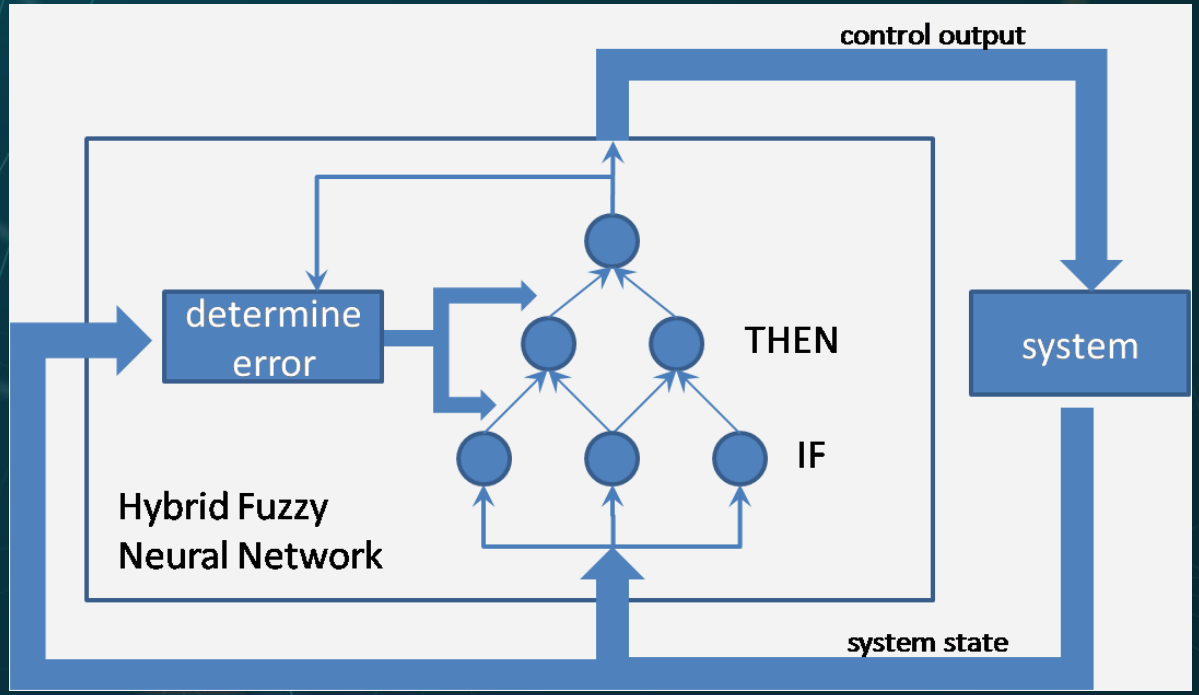
02 Decision making – Ability to build decision support systems is important

General Structure of Neuro-Fuzzy Model



Neuro-Fuzzy System

General Structure of Neuro-Fuzzy Model



Neuro-Fuzzy System

Structures of Neuro-Fuzzy model

01

Generic hybrid model

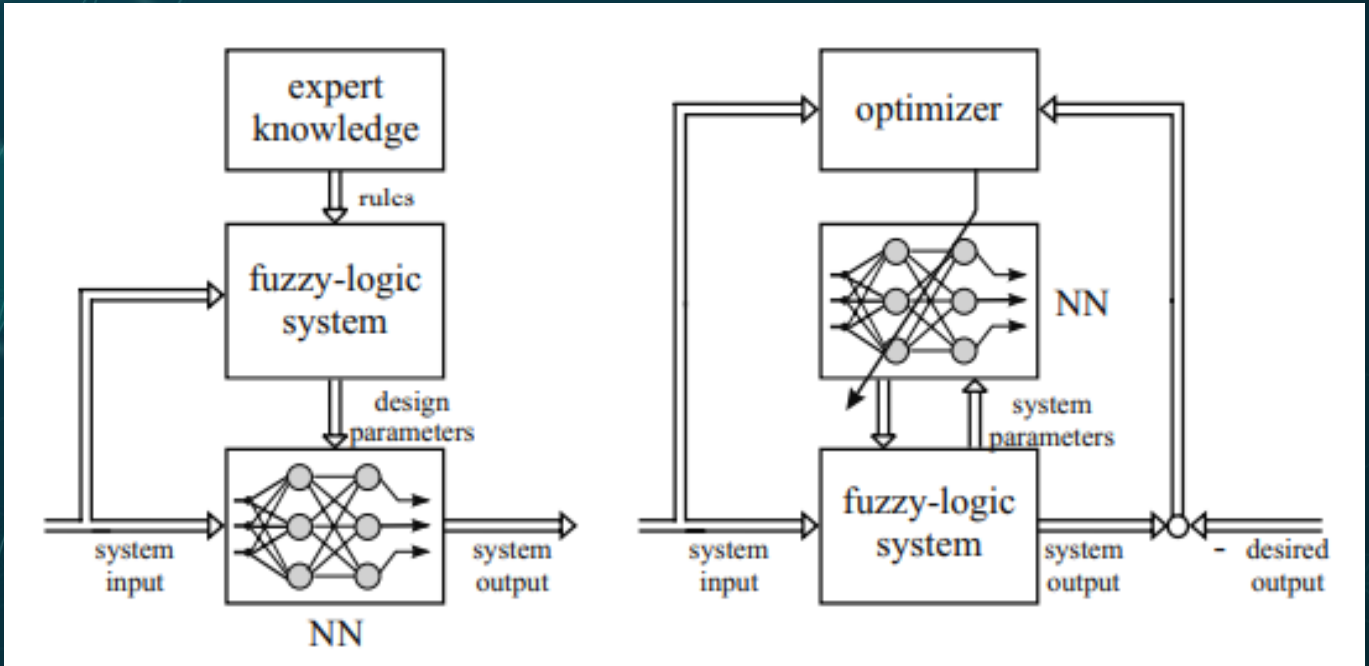
02

Mamdani and Singleton model

03

Takagi and Sugeno model

Generic Hybrid Neuro-Fuzzy Model

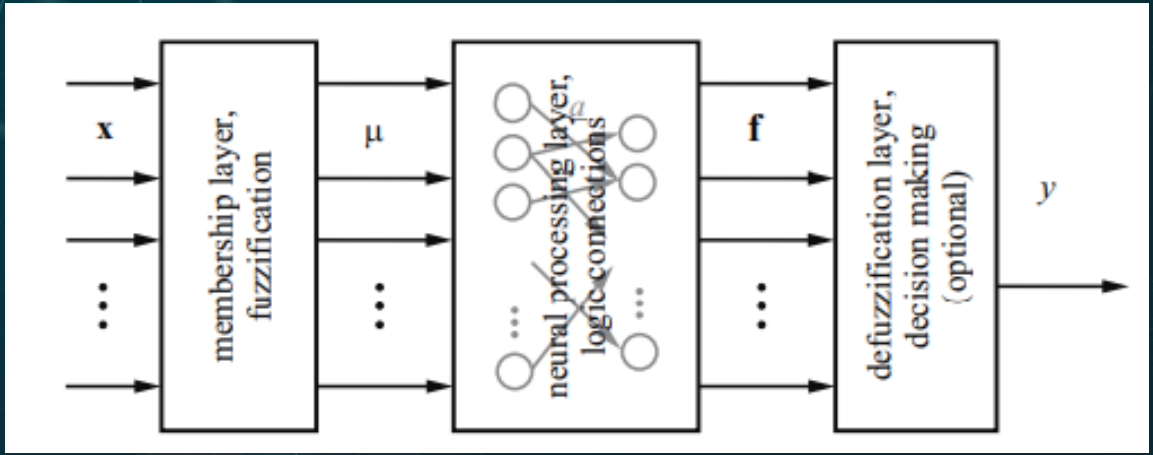


Serial and Parallel networks of general neuro-fuzzy systems

Generic Hybrid Neuro-Fuzzy Model

Hybrid neuro-fuzzy network consists of the same components as fuzzy inference engine

- Fuzzification layer
- Neural processing unit
- Defuzzification layer



Generic neuro-fuzzy structure (decision making)

Classification

- The classification problems maps the continuous, discrete or mixed input domains to the discrete output domain (0 and 1) - classes
- In the application of fault diagnosis, the inputs are
- ✓ Inputs (Symptoms– s_i)
- ✓ Outputs (fault measures - f_j)

R_l = IF ..... THEN class1

- In Classification, output fuzzy attributes do not exist.
- The maximum decision will directly act on activation of rule premises, the conclusion of the rule with strongest activation will be the classification statement

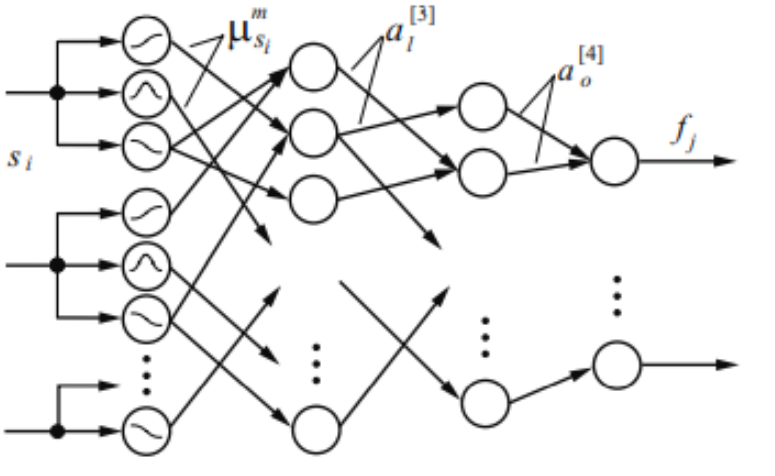
Regression

- A regression problem is the approximation of usually continuous valued one or higher dimensional function
- For 1-D, it refers to **y = f (x)** where y assumes any value in continuous output domain
- Regression neural network usually possess a defuzzification with output member functions or singletons and network elements that perform a weighted sum, which in case of singletons is equivalent to a center of gravity defuzzification

R_l = IF …… THEN y is large

Mamdani and singleton neuro-fuzzy Systems

- The typical hybrid neuro-fuzzy systems is the structure suggested by ***Fuzzy Adaptive Learning Control/Decision Network (FALCON)***



layer	1	2	3	4	5	
function	input **s**	fuzzifi-cation	AND	OR	defuzzi-fication	output **f**

Fuzzy Adaptive Learning Control/Decision Network (FALCON)

Mamdani and singleton neuro-fuzzy Systems

- The network structure maps Mamdani-type fuzzy rules like,

$$R_j : \text{IF } s_1 \text{ is } A_1 \text{ AND } s_2 \text{ is } A_2 \ldots \text{ AND } s_n \text{ is } A_n \text{ THEN yk is } B_1$$

- Triangular Membership functions are used in their ***Neuro-Fuzzy Classification System*** (NEFCLASS)
- Here, The inference involves Min-Max operators
- A different network structure called ***System Adaptive Rule Acquisition with Hebbian Learning*** (SARAH) which is also called as ANDOR neuron model.

Mamdani and singleton neuro-fuzzy Systems

- This represents both *t-norm* (parametric composite of bounded difference operator) and *t-conorm* (parametric composite of bounded sum operator)
- The resulting neuron resembles a standard sigmoidal perceptron
- The input is fuzzified by Gaussian membership functions in the antecedent layer of **SARAH** structure
- The rule activation scales the associated output singletons within the output layer
- This yields a Crisp output value

Takagi-Sugeno neuro-fuzzy Systems

- In Takagi-Sugeno structure the consequence is given as linear or nonlinear function of the inputs.

$$R_l: \text{IF } s_1 \text{ is } A_1 \text{ AND } s_2 \text{ is } A_2 \ldots \text{ AND } S_{ns} \text{ is } A_{na} \text{ THEN } f_j = F_{l,j}(s)$$

- The linear function in the consequence is,

$$R_l: \text{IF } s_1 \text{ is } A_1 \text{ AND } s_2 \text{ is } A_2 \ldots \text{ AND } s_{ns} \text{ is } A_{na} \text{ THEN } f_{l,j} = w_0 + w_1 s_1 + w_2 s_2 + \ldots + w_{ns} s_{ns}$$

- The overall output is then determined as the weighted sum over all consequences

Takagi-Sugeno neuro-fuzzy Systems

Learning Algorithms:

- The ***Adaptive-Network-based Fuzzy Inference System*** (ANFIS) is used.
- Inputs are used twice : Inputs of fuzzification layer and the inputs of consequences
- TS-fuzzy systems are used for the approximation of non-linear functions
- Adaptive-Network-based Fuzzy Inference System (ANFIS)

IF s1 is large THEN fault F2
ELSE IF s2 small THEN fault F1
ELSE fault F3

- **Local Linear Model Tree** (LOLIMOT) – Approximation of Non-linear systems and Pattern recognition

Case Study

FDI OF AN ELECTRO - PNEUMATIC VALVE ACTUATOR

- The valve considered for FDI is an electro-pneumatic flow controller in evaporization stage
- The model is then used to generate faulty/ fault-free data to evaluate the Neuro-fuzzy based fault isolation schemes

Conclusion

- Neuro-fuzzy systems not only have powerful approximation abilities for modelling unknown dynamic non-linear systems, but a high level description of the system can also be obtained.
- The transparent structure of NF is very useful to study the effect of faults on system characteristics

References

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THANK YOU